

Panel Data Part 2 and Formatting

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```
bookdown::render_book()
```


Chapter 1

Overview

We will cover the implementation the fixed effects (within) estimator and the random effects estimator. For Mitchell, we will cover the second half of Chapter 5 on formatting.

1. Fixed Effects (Within) Estimator
2. Random Effects Estimator
3. Formatting

Chapter 2

Fixed Effects (Within) Estimator

We will implement the fixed effects (within) estimator. We will need to use the *xtset* command to establish our unit of analysis dimension and our time dimension in our panel data. Our main stata command to implement a FD is *xtreg* with the option *fe*. It is important to include the option *fe*. If you forget, your model will be a random effects model.

2.1 Union Premium

Lesson: we can control for ability and preference for unionization to control for unobserved time-invariant confounders.

We want to estimate the union premium, and we will use a fixed effects (within) estimator to control for time-invariant heterogeneity, such as ability and performance.

$$\ln(wage_{i,t}) = \beta_0 + \beta_1 edu_{i,t} + \beta_2 exp_{i,t} + \beta_3 exp_{i,t}^2 + \beta_4 Black_i + \beta_7 South_{i,t} + \beta_8 married_{i,t} + union_{i,t} + a_t + a_i + \varepsilon_{i,t}$$

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use "wagepan.dta", clear
```

Set the Panel and estimate the Pooled OLS model

```
xtset nr year
eststo OLS: reg lwage c.edu exper expersq i.black i.south i.married i.union i.d8*
```

```
panel variable: nr (strongly balanced)
time variable: year, 1980 to 1987
delta: 1 unit
```

Source	SS	df	MS	Number of obs	=	4,360
Model	236.577196	14	16.8983712	F(14, 4345)	=	73.43
Residual	999.952446	4,345	.230138653	Prob > F	=	0.0000
				R-squared	=	0.1913
				Adj R-squared	=	0.1887
Total	1236.52964	4,359	.283672779	Root MSE	=	.47973

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0895815	.0051762	17.31	0.000	.0794334	.0997295
exper	.0665418	.0136722	4.87	0.000	.0397372	.0933463
expersq	-.0023832	.0008183	-2.91	0.004	-.0039875	-.0007789
1.black	-.1275978	.0236068	-5.41	0.000	-.1738791	-.0813165
1.south	-.0529354	.0155534	-3.40	0.001	-.0834281	-.0224428
1.married	.1122106	.0157149	7.14	0.000	.0814014	.1430198
1.union	.1802554	.0171339	10.52	0.000	.1466642	.2138466
1.d81	.058693	.0303147	1.94	0.053	-.0007393	.1181253
1.d82	.0638086	.0331714	1.92	0.054	-.0012242	.1288414
1.d83	.0633355	.0366129	1.73	0.084	-.0084444	.1351153
1.d84	.0915252	.0400387	2.29	0.022	.0130289	.1700216
1.d85	.1105424	.0432973	2.55	0.011	.0256575	.1954272
1.d86	.1437822	.0463663	3.10	0.002	.0528806	.2346837
1.d87	.1756699	.0493736	3.56	0.000	.0788725	.2724673
_cons	.1333638	.0774193	1.72	0.085	-.0184175	.285145

If we use FE or FD, we cannot assess race, education, or experience since they remain constant.

FE Within.

```
eststo Within: xtreg lwage c.edu exper expersq i.black i.south i.married i.union i.d8*
```

note: educ omitted because of collinearity

note: 1.black omitted because of collinearity

note: 1.d87 omitted because of collinearity

2.1. UNION PREMIUM

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```
Fixed-effects (within) regression      Number of obs   =    4,360
Group variable: nr                    Number of groups =     545

R-sq:                                Obs per group:
    within = 0.1815                      min =          8
    between = 0.0009                     avg  =         8.0
    overall = 0.0497                     max  =          8

corr(u_i, Xb) = -0.1739                F(11,3804)      =    76.71
                                           Prob > F        =    0.0000
```

	lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
	educ	0	(omitted)				
	exper	.1313318	.0098277	13.36	0.000	.1120638	.1505999
	expersq	-.0051475	.0007043	-7.31	0.000	-.0065284	-.0037666
	1.black	0	(omitted)				
	1.south	.1018493	.0479605	2.12	0.034	.0078186	.19588
	1.married	.0462057	.0183034	2.52	0.012	.0103203	.082091
	1.union	.0809394	.0193065	4.19	0.000	.0430874	.1187914
	1.d81	.0190821	.0203532	0.94	0.349	-.0208222	.0589864
	1.d82	-.0117214	.0202191	-0.58	0.562	-.0513627	.0279199
	1.d83	-.0425071	.0203126	-2.09	0.036	-.0823318	-.0026825
	1.d84	-.0378955	.0203069	-1.87	0.062	-.0777089	.0019179
	1.d85	-.0427598	.0202377	-2.11	0.035	-.0824377	-.0030819
	1.d86	-.0276303	.0203773	-1.36	0.175	-.0675818	.0123211
	1.d87	0	(omitted)				
	_cons	.9952994	.033587	29.63	0.000	.929449	1.06115
	sigma_u	.40924292					
	sigma_e	.35082825					
	rho	.57640294	(fraction of variance due to u_i)				

F test that all u_i=0: F(544, 3804) = 9.54 Prob > F = 0.0000

Compare with *esttab*

```
esttab OLS Within, mtitles se scalars(F r2) drop(0.d* 0.union 0.married)
```

	(1) OLS	(2) Within
educ	0.0896*** (0.00518)	0 (.)

exper	0.0665*** (0.0137)	0.131*** (0.00983)
expersq	-0.00238** (0.000818)	-0.00515*** (0.000704)
0.black	0 (.)	0 (.)
1.black	-0.128*** (0.0236)	0 (.)
0.south	0 (.)	0 (.)
1.south	-0.0529*** (0.0156)	0.102* (0.0480)
1.married	0.112*** (0.0157)	0.0462* (0.0183)
1.union	0.180*** (0.0171)	0.0809*** (0.0193)
1.d81	0.0587 (0.0303)	0.0191 (0.0204)
1.d82	0.0638 (0.0332)	-0.0117 (0.0202)
1.d83	0.0633 (0.0366)	-0.0425* (0.0203)
1.d84	0.0915* (0.0400)	-0.0379 (0.0203)
1.d85	0.111* (0.0433)	-0.0428* (0.0202)
1.d86	0.144** (0.0464)	-0.0276 (0.0204)
1.d87	0.176*** (0.0494)	0 (.)

_cons	0.133 (0.0774)	0.995*** (0.0336)

N	4360	4360
F	73.43	76.71
r2	0.191	0.182

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

After controlling for time-invariant individual fixed effects the Pooled OLS is seen to be upward biased. The union wage premium is about 19.7% with the Pooled OLS model, while the union wage premium is estimated to be 8.4% after accounting for individual fixed effects.

Pooled OLS Wage Premium Estimate

```
display (exp(_b[1.union])-1)*100
```

19.752317

FE Within Wage Premium Estimate

```
display (exp(_b[1.union])-1)*100
```

8.4305206

Plot the Coefficients of Interest *Credit:* John Kane: Making Regression Coefficients Plots in Stata

```
quietly reg lwage c.edu exper expersq i.black i.south i.married i.union i.d8*
estimates store pooled
quietly xtreg lwage c.edu i.black i.south i.married i.union i.d8*, fe
estimates store fe
coefplot (pooled, label("{bf:Pooled OLS}") mcolor(midblue) mlcolor(cyan) ///
ciopts(lcolor(magenta midblue))) /// options for first group
(fe, label("{bf: Within}") mcolor(green) mlcolor(lime) ///
ciopts(lcolor(lime green))), /// options for second group
title(Union and Marriage Wage Premiums) ///
drop(_cons 1.d* 0.black 0.south) ///
xline(0, lcolor(red) lwidth(medium)) scheme(jet_white) ///
xtitle("{bf: Effect of Union}") ///
graphregion(margin(smaller)) ///
coeflabels(educ="Education" exper="Experience" expersq="Experience Squared") ///
```

```

1.black="Black" 1.south="South" 1.married="Married" ///
1.union="Union") ///
msize(large) mcolor(%85) mlwidth(medium) msymbol(circle) /// marker options
levels(95 90) ciopts(lwidth(medthick thick) recast(rspike rcap)) ///ci options f
legend(ring(1) col(1) pos(3) size(medsmall))
graph export "/Users/Sam/Desktop/Econ 645/Stata/week4_union_wage_premium.png", r

```

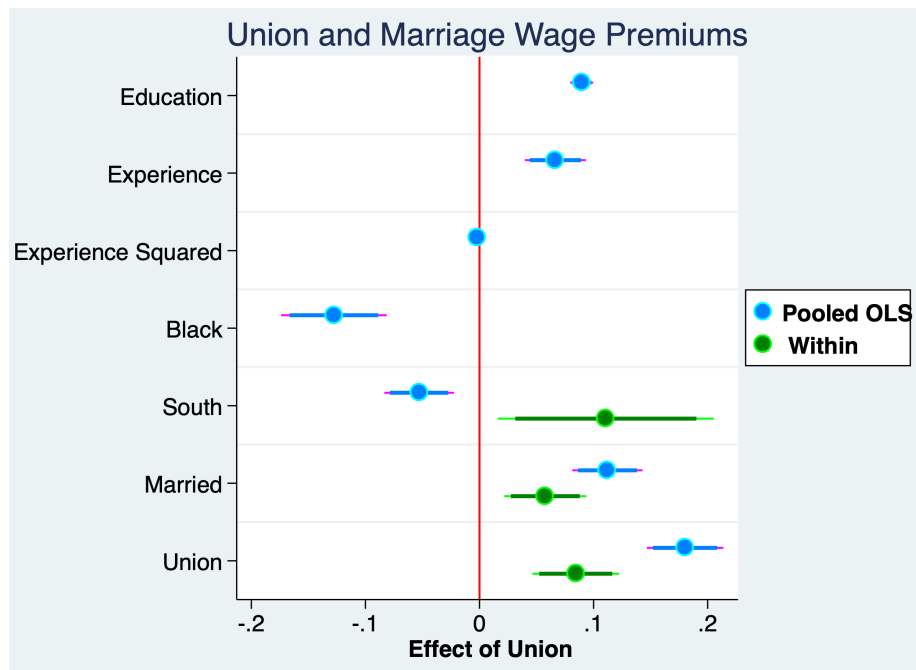


Figure 2.1: Plot the Coefficients of Interest

2.2 Has returns to education changed over time

Lesson: We can interact time binaries with continuous time-invariant data to see if returns to education have changed over time<

With fixed effects or first differencing, we cannot assess time-invariant variables. Variables that do not vary over time, such as sex, race, or education (assuming) education is static. But, if we interact education with time binaries, we can assess whether returns to education have increased over time.

We can test to see if returns to education are constant over time.

Vella and Verbeek (1998) estimate to see if the returns to education have change over time. We have some variables that are not time-invariant, such as union

status and marital status. Experience does growth but it grows at a constant rate. We have a few variable that do not (or we would expect not to change), such as race and education (for older workers).

We use the natural log of wages, which has nice properties, such as being more normally distributed and providing elasticities. It also can take care of inflation when we add time period binaries.

Set up the Panel

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "wagepan.dta", clear
xtset nr year
```

Pooled OLS

```
eststo OLS: reg lwage c.edu##i.d8* exper expersq i.married i.union
```

Source	SS	df	MS	Number of obs	=	4,360
Model	225.412805	19	11.8638318	F(19, 4340)	=	50.92
Residual	1011.11684	4,340	.23297623	Prob > F	=	0.0000
				R-squared	=	0.1823
				Adj R-squared	=	0.1787
Total	1236.52964	4,359	.283672779	Root MSE	=	.48268

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.081673	.0125675	6.50	0.000	.0570343	.1063117
1.d81	-.0356958	.199359	-0.18	0.858	-.4265413	.3551497
1.d82	-.0315288	.1998095	-0.16	0.875	-.4232575	.3602
1.d83	-.0342801	.2007839	-0.17	0.864	-.4279192	.3593589
1.d84	.0242933	.2025167	0.12	0.905	-.3727429	.4213294
1.d85	.0058838	.2052301	0.03	0.977	-.3964719	.4082395
1.d86	.0251586	.2092184	0.12	0.904	-.3850164	.4353336
1.d87	.0372565	.2148364	0.17	0.862	-.3839326	.4584456
d81#c.educ						
1	.0084448	.0167792	0.50	0.615	-.024451	.0413407
d82#c.educ						
1	.0088899	.0168742	0.53	0.598	-.0241921	.041972
d83#c.educ						
1	.0093544	.0170326	0.55	0.583	-.0240381	.042747

d84#c.educ							
1		.0070671	.0172551	0.41	0.682	-.0267617	.0408958
d85#c.educ							
1		.0104027	.0175306	0.59	0.553	-.0239662	.0447716
d86#c.educ							
1		.0116562	.0178614	0.65	0.514	-.0233613	.0466737
d87#c.educ							
1		.0134166	.0182525	0.74	0.462	-.0223676	.0492008
exper		.0568876	.0154436	3.68	0.000	.0266102	.087165
expersq		-.001919	.0009455	-2.03	0.042	-.0037726	-.0000654
1.married		.1229473	.0155752	7.89	0.000	.0924119	.1534827
1.union		.1720565	.0171378	10.04	0.000	.1384575	.2056554
_cons		.2175863	.1641736	1.33	0.185	-.1042777	.5394503

Fixed Effects (Within)

eststo Within: `xtreg lwage c.edu##i.d8* i.married i.union, fe`

note: educ omitted because of collinearity

Fixed-effects (within) regression	Number of obs	=	4,360
Group variable: nr	Number of groups	=	545
R-sq:	Obs per group:		
within = 0.1708	min =		8
between = 0.1900	avg =		8.0
overall = 0.1325	max =		8
	F(16,3799)	=	48.91
corr(u_i, Xb) = 0.0991	Prob > F	=	0.0000

lwage		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
educ		0	(omitted)			
1.d81		-.0224158	.1458885	-0.15	0.878	-.3084431 .2636114
1.d82		-.0057611	.1458558	-0.04	0.968	-.2917243 .2802021
1.d83		.0104297	.1458579	0.07	0.943	-.2755377 .2963971
1.d84		.0843743	.1458518	0.58	0.563	-.2015811 .3703297
1.d85		.0497253	.1458602	0.34	0.733	-.2362465 .3356971

1.d86		.0656064	.1458917	0.45	0.653	-.2204273	.3516401
1.d87		.0904448	.1458505	0.62	0.535	-.195508	.3763977
d81#c.educ							
1		.0115854	.0122625	0.94	0.345	-.0124562	.0356271
d82#c.educ							
1		.0147905	.0122635	1.21	0.228	-.0092533	.0388342
d83#c.educ							
1		.0171182	.0122633	1.40	0.163	-.0069251	.0411615
d84#c.educ							
1		.0165839	.0122657	1.35	0.176	-.007464	.0406319
d85#c.educ							
1		.0237085	.0122738	1.93	0.053	-.0003554	.0477725
d86#c.educ							
1		.0274123	.012274	2.23	0.026	.0033481	.0514765
d87#c.educ							
1		.0304332	.0122723	2.48	0.013	.0063722	.0544942
1.married		.0548205	.0184126	2.98	0.003	.018721	.09092
1.union		.0829785	.0194461	4.27	0.000	.0448527	.1211042
_cons		1.362459	.0162385	83.90	0.000	1.330622	1.394296
-----+-----							
sigma_u		.37264193					
sigma_e		.35335713					
rho		.52654439	(fraction of variance due to u_i)				

F test that all u_i=0: F(544, 3799) = 8.09					Prob > F = 0.0000		

If we use FE or FD, we cannot assess race, education, or experience since they remain constant, but we can include dummy interactions.

These are changes in the returns to education compared to the base year of 1980
And only 1987 * *education* and 1986 * *education* appear to be insignificant

Compare

```
esttab OLS Within, mtitles se scalars(F r2) drop(0.d* 0.union 0.married)
```

(1)	(2)
OLS	Within

educ	0.0817*** (0.0126)	0 (.)
1.d81	-0.0357 (0.199)	-0.0224 (0.146)
1.d82	-0.0315 (0.200)	-0.00576 (0.146)
1.d83	-0.0343 (0.201)	0.0104 (0.146)
1.d84	0.0243 (0.203)	0.0844 (0.146)
1.d85	0.00588 (0.205)	0.0497 (0.146)
1.d86	0.0252 (0.209)	0.0656 (0.146)
1.d87	0.0373 (0.215)	0.0904 (0.146)
1.d81#c.educ	0.00844 (0.0168)	0.0116 (0.0123)
1.d82#c.educ	0.00889 (0.0169)	0.0148 (0.0123)
1.d83#c.educ	0.00935 (0.0170)	0.0171 (0.0123)
1.d84#c.educ	0.00707 (0.0173)	0.0166 (0.0123)
1.d85#c.educ	0.0104 (0.0175)	0.0237 (0.0123)
1.d86#c.educ	0.0117 (0.0179)	0.0274* (0.0123)
1.d87#c.educ	0.0134 (0.0183)	0.0304* (0.0123)

exper	0.0569*** (0.0154)	
expersq	-0.00192* (0.000945)	
1.married	0.123*** (0.0156)	0.0548** (0.0184)
1.union	0.172*** (0.0171)	0.0830*** (0.0194)
_cons	0.218 (0.164)	1.362*** (0.0162)

N	4360	4360
F	50.92	48.91
r2	0.182	0.171

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Returns to education have increased by about 3.1% between 1987 and 1980.

```
display (exp(0.0304)-1)*100
```

3.0866798

Plot the Coefficients

```
quietly reg lwage c.edu##i.d8* exper expersq i.married i.union
estimates store pooled
quietly xtreg lwage c.edu##i.d8* i.married i.union, fe
estimates store fe
coefplot ///
    (pooled, label("{bf:Pooled OLS}") mcolor(midblue) mlcolor(cyan) ///
        ciopts(lcolor(magenta midblue))) /// options for first group
    (fe, label("{bf: Within}") mcolor(green) mlcolor(lime) ///
        ciopts(lcolor(lime green))), /// options for second group
title("Change in Return to Education") ///
keep(educ 1.d81#c.educ 1.d82#c.educ 1.d83#c.educ 1.d84#c.educ ///
    1.d85#c.educ 1.d86#c.educ 1.d87#c.educ) ///
xline(0, lcolor(red) lwidth(medium)) scheme(jet_white) ///
xtitle("{bf: Coefficients}") ///
```

```

graphregion(margin(small)) ///
coeflabels(educ="Education" 1.d81#c.educ="1981 Returns" ///
1.d82#c.educ="1982 Returns" 1.d83#c.educ="1983 Returns" ///
1.d84#c.educ="1984 Returns" 1.d85#c.educ="1985 Returns" ///
1.d86#c.educ="1986 Returns" 1.d87#c.educ="1987 Returns") ///
msize(large) mcolor(%85) mlwidth(medium) msymbol(circle) /// marker options
levels(95 90) ciopts(lwidth(medthick thick) recast(rspike rcap)) ///ci options f
legend(ring(1) col(1) pos(3) size(medsmall))
graph export "/Users/Sam/Desktop/Econ 645/Stata/week4_edu_returns.png", replace

```

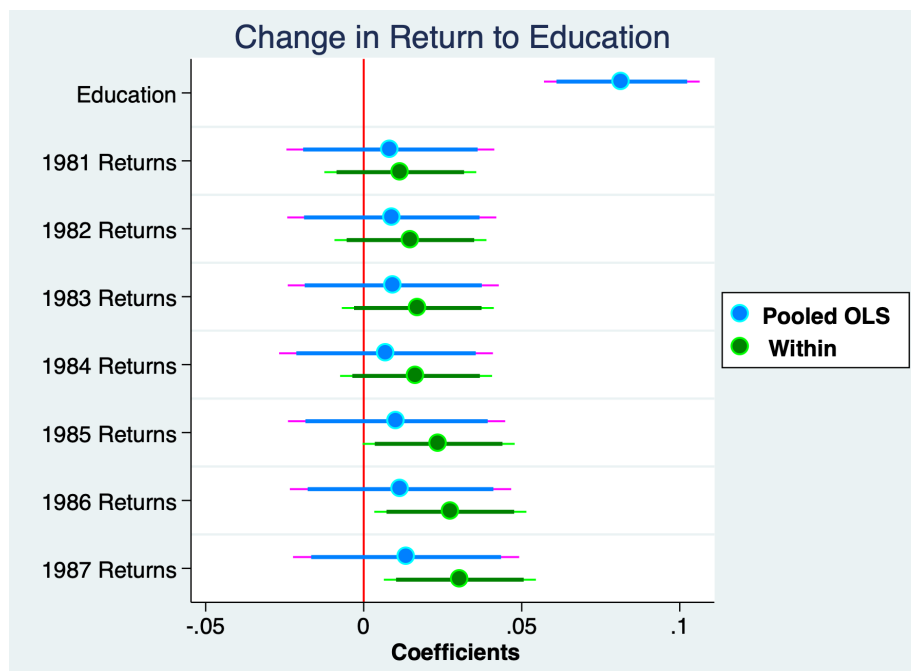


Figure 2.2: Plot the Coefficients of Interest

Test for Serial Correlation

Test for Serial Correlation. Use the option *residuals* or *resid* to get post-estimation residuals. If you don't specify *resid*, Stata will return $(\hat{u}_t)$ instead of $(\hat{u}_{t-1})$

Our null hypothesis is that there is no serial correlation or the coefficient on our lagged residuals is zero. We'll regress u on lag of u AR(1) model without a constant

```
quietly xtreg lwage c.edu##i.d8* i.married i.union, fe
predict u, resid
reg u l.u, noconst
```

Source	SS	df	MS	Number of obs	=	3,815
Model	332.737937	1	332.737937	F(1, 3814)	=	2181.06
Residual	581.856111	3,814	.152557973	Prob > F	=	0.0000
				R-squared	=	0.3638
				Adj R-squared	=	0.3636
Total	914.594048	3,815	.239736317	Root MSE	=	.39059

u	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
u						
L1.	.5857228	.0125418	46.70	0.000	.5611336	.610312

We can see that we have positive serial correlation since the coefficient on our lagged residual is positive and statistically significant. We will need to cluster our standard errors to account for the positive serial correlation.

Dealing with Heteroskedasticity and Serial Correlation

For heteroskedasticity, we will need to use heteroskedasticity-robust standard errors by using the *robust* option.

For serial correlation, we will need to cluster our standard errors. We will cluster the standard errors at the unit of analysis level.

```
eststo nocluster: quietly xtreg lwage c.edu##i.d8* i.married i.union, fe
eststo clustered: quietly xtreg lwage c.edu##i.d8* i.married i.union, fe robust cluster(nr)

esttab nocluster clustered, mtitle ("FE" "FE Clustered") drop(0.*) se
```

	(1) FE	(2) FE Clustered
educ	0 (.)	0 (.)
1.d81	-0.0224 (0.146)	-0.0224 (0.144)

1.d82	-0.00576 (0.146)	-0.00576 (0.139)
1.d83	0.0104 (0.146)	0.0104 (0.154)
1.d84	0.0844 (0.146)	0.0844 (0.159)
1.d85	0.0497 (0.146)	0.0497 (0.157)
1.d86	0.0656 (0.146)	0.0656 (0.171)
1.d87	0.0904 (0.146)	0.0904 (0.157)
1.d81#c.educ	0.0116 (0.0123)	0.0116 (0.0122)
1.d82#c.educ	0.0148 (0.0123)	0.0148 (0.0118)
1.d83#c.educ	0.0171 (0.0123)	0.0171 (0.0131)
1.d84#c.educ	0.0166 (0.0123)	0.0166 (0.0138)
1.d85#c.educ	0.0237 (0.0123)	0.0237 (0.0136)
1.d86#c.educ	0.0274* (0.0123)	0.0274 (0.0147)
1.d87#c.educ	0.0304* (0.0123)	0.0304* (0.0135)
1.married	0.0548** (0.0184)	0.0548* (0.0212)
1.union	0.0830*** (0.0194)	0.0830*** (0.0230)
_cons	1.362***	1.362***

	(0.0162)	(0.0203)

N	4360	4360

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

2.3 Rental Prices

Exercise 1: Compare Pooled OLS, First Difference, and Fixed Effects Within

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "rental.dta", clear
```

Get the data here: rental.dta

The data on rental prices and other variables in college towns from 1980 to 1990. Do more students affect the prices? The general model with unobserved fixed effects is

$$\ln(\text{rent}_{i,t}) = \beta_0 + \delta_0 y90_t + \beta_1 \ln(\text{pop}_{i,t}) + \beta_2 \ln(\text{avginc}_{i,t}) + \beta_4 \text{pctstu}_{i,t} + a_t + a_i + \varepsilon_{i,t}$$

Where *pop* is city population, *avginc* is average income, *pctstu* is the student percent of the population, and *rent* is the nominal rental prices

1. Estimate a Pooled OLS. What does the estimate on y90 tell you?
2. Are there concerns with the standard errors in the Pooled OLS?
3. Use a First difference model. Does the coefficient on b3 change?
4. Use a FE Within model. Are the results the same as the FD model?

Set the Panel

```
xtset city year, delta(10)
```

```

panel variable:  city (strongly balanced)
time variable:  year, 80 to 90
delta:          10 units
```

Pooled OLS

```
reg lrent i.y90 lpop lavginc pctstu, robust
```

Linear regression	Number of obs	=	128
	F(4, 123)	=	223.26
	Prob > F	=	0.0000
	R-squared	=	0.8613
	Root MSE	=	.12592

	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
lrent						
1.y90	.2622267	.0579584	4.52	0.000	.1475017	.3769517
lpop	.0406863	.0223732	1.82	0.071	-.0036	.0849726
lavginc	.5714461	.0989016	5.78	0.000	.3756765	.7672157
pctstu	.0050436	.0011488	4.39	0.000	.0027696	.0073176
_cons	-.5688069	.8506229	-0.67	0.505	-2.252563	1.114949

First Difference

```
reg d.lrent i.y90 d.lpop d.lavginc d.pctstu
```

note: 1.y90 omitted because of collinearity

Source	SS	df	MS	Number of obs	=	64
				F(3, 60)	=	9.51
Model	.231738668	3	.077246223	Prob > F	=	0.0000
Residual	.487362198	60	.008122703	R-squared	=	0.3223
				Adj R-squared	=	0.2884
Total	.719100867	63	.011414299	Root MSE	=	.09013

	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D.lrent						
1.y90	0 (omitted)					
lpop						
D1.	.0722456	.0883426	0.82	0.417	-.104466	.2489571
lavginc						
D1.	.3099605	.0664771	4.66	0.000	.1769865	.4429346
pctstu						
D1.	.0112033	.0041319	2.71	0.009	.0029382	.0194684
_cons	.3855214	.0368245	10.47	0.000	.3118615	.4591813

Fixed Effects

`xtreg lrent i.y90 lpop lavginc pctstu, fe`

```

Fixed-effects (within) regression           Number of obs   =       128
Group variable: city                       Number of groups =        64

R-sq:                                     Obs per group:
    within = 0.9765                        min =           2
    between = 0.2173                       avg  =          2.0
    overall = 0.7597                       max  =           2

                                           F(4,60)         =      624.15
corr(u_i, Xb) = -0.1297                   Prob > F         =      0.0000

```

	lrent	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
	1.y90	.3855214	.0368245	10.47	0.000	.3118615	.4591813
	lpop	.0722456	.0883426	0.82	0.417	-.104466	.2489571
	lavginc	.3099605	.0664771	4.66	0.000	.1769865	.4429346
	pctstu	.0112033	.0041319	2.71	0.009	.0029382	.0194684
	_cons	1.409384	1.167238	1.21	0.232	-.9254394	3.744208
	sigma_u	.15905877					
	sigma_e	.06372873					
	rho	.8616755	(fraction of variance due to u_i)				

F test that all u_i=0: F(63, 60) = 6.67 Prob > F = 0.0000

Chapter 3

Random Effects

We will implement our Random Effects estimator in Stata. It is similar to fixed effects. We use *xtreg*, but we use the *re* option.

Remember that we need to test our main random effects assumption to see if a random effects estimate is appropriate:

$$Cov(x_{i,j}, a_i) = 0$$

3.1 Returns to Marriage for Men

Lesson: We can test to see if random effects is an appropriate assumption.

We'll use three methods to estimate the returns to marriage for men: Pooled OLS, Fixed Effects (Within), and Random Effects. We cannot estimate the coefficients for Black and Latinos.

We can use the wagepan data again to estimate the returns to marriage for men. We will compare the Pooled OLS, FE (Within), and Random Effects estimates

Set Panel

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use "wagepan.dta", clear  
xtset nr year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
panel variable:  nr (strongly balanced)  
time variable:   year, 1980 to 1987  
delta:          1 unit
```

Pooled OLS

```
reg lwage educ i.black i.hisp exper expersq married union i.d8*
eststo m1: quietly reg lwage educ i.black i.hisp exper expersq married union i.d8*
```

Source	SS	df	MS	Number of obs	=	4,360
Model	234.048277	14	16.7177341	F(14, 4345)	=	72.46
Residual	1002.48136	4,345	.230720682	Prob > F	=	0.0000
				R-squared	=	0.1893
				Adj R-squared	=	0.1867
Total	1236.52964	4,359	.283672779	Root MSE	=	.48033

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0913498	.0052374	17.44	0.000	.0810819	.1016177
1.black	-.1392342	.0235796	-5.90	0.000	-.1854622	-.0930062
1.hisp	.0160195	.0207971	0.77	0.441	-.0247535	.0567925
exper	.0672345	.0136948	4.91	0.000	.0403856	.0940834
expersq	-.0024117	.00082	-2.94	0.003	-.0040192	-.0008042
married	.1082529	.0156894	6.90	0.000	.0774937	.1390122
union	.1824613	.0171568	10.63	0.000	.1488253	.2160973
1.d81	.05832	.0303536	1.92	0.055	-.0011886	.1178286
1.d82	.0627744	.0332141	1.89	0.059	-.0023421	.1278909
1.d83	.0620117	.0366601	1.69	0.091	-.0098608	.1338843
1.d84	.0904672	.0400907	2.26	0.024	.011869	.1690654
1.d85	.1092463	.0433525	2.52	0.012	.0242533	.1942393
1.d86	.1419596	.046423	3.06	0.002	.0509469	.2329723
1.d87	.1738334	.049433	3.52	0.000	.0769194	.2707474
_cons	.0920558	.0782701	1.18	0.240	-.0613935	.2455051

Pooled OLS shows a marriage premium of:

11.432957

The Pooled OLS data are likely upward biased - self-selection into marriage and we will have positive serial correlation so we really should cluster our standard errors by the group id.

Fixed Effects (Within)

```
xtreg lwage educ i.black i.hisp exper expersq married union i.d8*, fe
```

```
note: educ omitted because of collinearity
note: 1.black omitted because of collinearity
note: 1.hisp omitted because of collinearity
note: 1.d87 omitted because of collinearity
```

Fixed-effects (within) regression	Number of obs	=	4,360
Group variable: nr	Number of groups	=	545

R-sq:		Obs per group:	
within = 0.1806		min =	8
between = 0.0005		avg =	8.0
overall = 0.0635		max =	8

	F(10,3805)	=	83.85
corr(u_i, Xb) = -0.1212	Prob > F	=	0.0000

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	0	(omitted)				
1.black	0	(omitted)				
1.hisp	0	(omitted)				
exper	.1321464	.0098247	13.45	0.000	.1128842	.1514087
expersq	-.0051855	.0007044	-7.36	0.000	-.0065666	-.0038044
married	.0466804	.0183104	2.55	0.011	.0107811	.0825796
union	.0800019	.0193103	4.14	0.000	.0421423	.1178614
1.d81	.0190448	.0203626	0.94	0.350	-.0208779	.0589674
1.d82	-.011322	.0202275	-0.56	0.576	-.0509798	.0283359
1.d83	-.0419955	.0203205	-2.07	0.039	-.0818357	-.0021553
1.d84	-.0384709	.0203144	-1.89	0.058	-.0782991	.0013573
1.d85	-.0432498	.0202458	-2.14	0.033	-.0829434	-.0035562
1.d86	-.0273819	.0203863	-1.34	0.179	-.0673511	.0125872
1.d87	0	(omitted)				
_cons	1.02764	.0299499	34.31	0.000	.9689201	1.086359
sigma_u	.4009279					
sigma_e	.35099001					
rho	.56612236	(fraction of variance due to u_i)				

F test that all u i=0: F(544, 3805) = 9.64 Prob > F = 0.0000

Fixed Effects model shows a marriage premium of:

4.7787041

We use estimates store to store our FE (Within) estimates to compare

```
estimates store femodel
```

```
eststo m2: quietly xtreg lwage educ i.black i.hisp exper expersq married union i.d8*,
```

Random Effects

We can use the theta option to find the lambda-hat GLS transformaton <https://www.stata.com/manuals/xtxtreg.pdf>

```
xtreg lwage educ i.black i.hisp exper expersq married union i.d8*, re theta
```

```
eststo m3: quietly xtreg lwage educ i.black i.hisp exper expersq married union i.d8*,
```

Random-effects GLS regression	Number of obs	=	4,360
Group variable: nr	Number of groups	=	545
R-sq:	Obs per group:		
within = 0.1799	min =		8
between = 0.1860	avg =		8.0
overall = 0.1830	max =		8
corr(u_i, X) = 0 (assumed)	Wald chi2(14)	=	957.77
theta = .64291089	Prob > chi2	=	0.0000

lwage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
educ	.0918763	.0106597	8.62	0.000	.0709836	.1127689
1.black	-.1393767	.0477228	-2.92	0.003	-.2329117	-.0458417
1.hisp	.0217317	.0426063	0.51	0.610	-.0617751	.1052385
exper	.1057545	.0153668	6.88	0.000	.0756361	.1358729
expersq	-.0047239	.0006895	-6.85	0.000	-.0060753	-.0033726
married	.063986	.0167742	3.81	0.000	.0311091	.0968629
union	.1061344	.0178539	5.94	0.000	.0711415	.1411273
1.d81	.040462	.0246946	1.64	0.101	-.0079385	.0888626
1.d82	.0309212	.0323416	0.96	0.339	-.0324672	.0943096
1.d83	.0202806	.041582	0.49	0.626	-.0612186	.1017798
1.d84	.0431187	.0513163	0.84	0.401	-.0574595	.1436969
1.d85	.0578155	.0612323	0.94	0.345	-.0621977	.1778286
1.d86	.0919476	.0712293	1.29	0.197	-.0476592	.2315544
1.d87	.1349289	.0813135	1.66	0.097	-.0244427	.2943005
_cons	.0235864	.1506683	0.16	0.876	-.271718	.3188907
sigma_u	.32460315					
sigma_e	.35099001					
rho	.46100216	(fraction of variance due to u_i)				

The Random Effects model shows a marriage premium of:

6.6077497

Our λ -hat is 0.643, which means it is closer to the FE estimator than the Pooled OLS estimator.

Hausman Test

We will use the *hausman* command to test the main random effects assumption $Cov(x_{i,j}, a_i) = 0$. First we specify the command *hausman*, then we input the two models. We stored our fixed effects model estimates into **femodel**. We will get our latest estimates from the random effects model with `."`. We will also use the option *sigmamore*.

```
hausman femodel ., sigmamore
```

Note: the rank of the differenced variance matrix (5) does not equal the number of coefficients being tested (10); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

---- Coefficients ----				
	(b)	(B)	(b-B)	sqrt(diag(V_b-V_B))
	femodel	.	Difference	S.E.
exper	.1321464	.1057545	.0263919	.
expersq	-.0051855	-.0047239	-.0004616	.0001533
married	.0466804	.063986	-.0173057	.0074632
union	.0800019	.1061344	-.0261326	.0074922
1.d81	.0190448	.040462	-.0214172	.
1.d82	-.011322	.0309212	-.0422431	.
1.d83	-.0419955	.0202806	-.0622762	.
1.d84	-.0384709	.0431187	-.0815896	.
1.d85	-.0432498	.0578155	-.1010653	.
1.d86	-.0273819	.0919476	-.1193295	.

b = consistent under H_0 and H_a ; obtained from xtreg
 B = inconsistent under H_a , efficient under H_0 ; obtained from xtreg

Test: Ho: difference in coefficients not systematic

```
chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
          =      26.22
Prob>chi2 =      0.0001
(V_b-V_B is not positive definite)
```

We reject the null hypothesis and a_i is correlated with the explanatory variables, so the random effects model is likely not appropriate.

```
esttab m1 m2 m3, drop(0.black 0.hisp 0.d8*) mtitles("Pooled OLS" "Within Model" "RE Model")
display (exp(.108)-1)*100
display (exp(.0467)-1)*100
display (exp(.064)-1)*100
```

	(1) Pooled OLS	(2) Within Model	(3) RE Model
educ	0.0913*** (17.44)	0 (.)	0.0919*** (8.62)
1.black	-0.139*** (-5.90)	0 (.)	-0.139** (-2.92)
1.hisp	0.0160 (0.77)	0 (.)	0.0217 (0.51)
exper	0.0672*** (4.91)	0.132*** (13.45)	0.106*** (6.88)
expersq	-0.00241** (-2.94)	-0.00519*** (-7.36)	-0.00472*** (-6.85)
married	0.108*** (6.90)	0.0467* (2.55)	0.0640*** (3.81)
union	0.182*** (10.63)	0.0800*** (4.14)	0.106*** (5.94)
1.d81	0.0583 (1.92)	0.0190 (0.94)	0.0405 (1.64)
1.d82	0.0628 (1.89)	-0.0113 (-0.56)	0.0309 (0.96)

1.d83	0.0620 (1.69)	-0.0420* (-2.07)	0.0203 (0.49)
1.d84	0.0905* (2.26)	-0.0385 (-1.89)	0.0431 (0.84)
1.d85	0.109* (2.52)	-0.0432* (-2.14)	0.0578 (0.94)
1.d86	0.142** (3.06)	-0.0274 (-1.34)	0.0919 (1.29)
1.d87	0.174*** (3.52)	0 (.)	0.135 (1.66)
_cons	0.0921 (1.18)	1.028*** (34.31)	0.0236 (0.16)

N	4360	4360	4360

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

11.404775

4.780762

6.6092399

We can see that the marriage premium falls from 11.4% in Pooled OLS to 4.8% in Fixed Effects. If we didn't reject our RE model, it would have been 6.6%.

The difference between the 11.4% Pooled OLS and the 4.8% in the Within Model might come from self-selection in marriage (they would have made more money even if they weren't married), and employers paying married men more if marriage is a sign of stability. But, we cannot distinguish these two hypotheses with this research design.

Plot the Coefficients

```
quietly reg lwage educ i.black i.hisp exper expersq married union i.d8*
estimates store pooled
quietly xtreg lwage educ i.black i.hisp exper expersq married union i.d8*, fe
```

```

estimates store fe
quietly xtreg lwage educ i.black i.hisp exper expersq married union i.d8*, re theta
estimates store re
coefplot ///
    (pooled, label("{bf:Pooled OLS}") mcolor(midblue) mlcolor(cyan) ///
      ciopts(lcolor(magenta midblue))) /// options for first group
    (fe, label("{bf: Within}") mcolor(green) mlcolor(lime) ///
      ciopts(lcolor(lime green))) /// options for second group
    (re, label("{bf: Random Effects}") mcolor(yellow) mlcolor(gold) ///
      ciopts(lcolor(gold yellow))), /// options for third group
    title("Returns to Marriage for Men") ///
    keep(married) ///
    xline(0, lcolor(red) lpattern(dash) lwidth(medium)) scheme(jet_white) ///
    xtitle("{bf: Coefficients}") ///
    graphregion(margin(small)) ///
    coelabels(married="Married") ///
    msize(large) mcolor(%85) mlwidth(medium) msymbol(circle) /// marker options
    levels(95 90) ciopts(lwidth(medthick thick) recast(rspike rcap)) /// ci options f
    legend(ring(1) col(1) pos(3) size(medsmall))
graph export "/Users/Sam/Desktop/Econ 645/Stata/week4_married_returns.png", replace

```

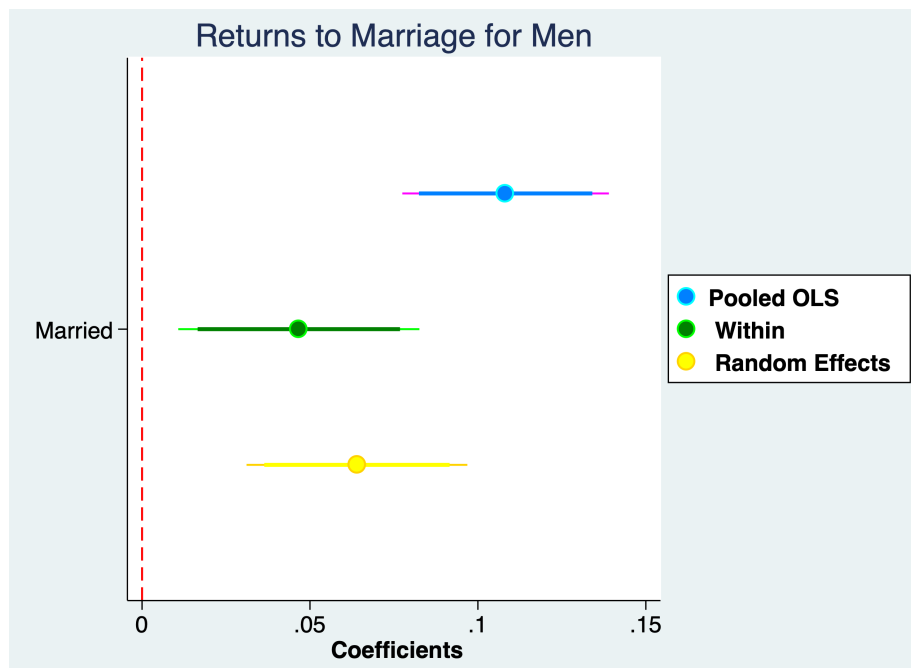


Figure 3.1: Plot the Coefficients of Interest

3.2 Airline prices and market concentration

We will assess concentration of airline on airfare. Our model:

$$\ln(fare_{i,t}) = \beta_0 + \beta_1 concn_{i,t} + \beta_2 \ln(dist_{i,t}) + \beta_3 (\ln(dist))^2 + a_i + a_t + \varepsilon_{i,t}$$

Estimate the Pooled OLS with time binaries

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "airfare.dta", clear
reg lfare concn ldist ldistsq i.y99 i.y00
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

Source	SS	df	MS	Number of obs	=	4,596
Model	355.197587	5	71.0395174	F(5, 4590)	=	627.18
Residual	519.896787	4,590	.113267274	Prob > F	=	0.0000
				R-squared	=	0.4059
				Adj R-squared	=	0.4052
Total	875.094374	4,595	.190444913	Root MSE	=	.33655

lfare	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
concn	.3609856	.0300677	12.01	0.000	.3020384	.4199327
ldist	-.9018838	.1282905	-7.03	0.000	-1.153395	-.6503727
ldistsq	.1030514	.0097268	10.59	0.000	.0839822	.1221207
1.y99	.0272979	.0121656	2.24	0.025	.0034475	.0511483
1.y00	.0893211	.012169	7.34	0.000	.0654641	.1131782
_cons	6.219743	.4206248	14.79	0.000	5.395116	7.04437

What is the associated change in airfare with a 10-percentage point increase in market concentration?

```
sum concn
```

Variable	Obs	Mean	Std. Dev.	Min	Max
concn	4,596	.6101149	.196435	.1605	1

What does the quadratic on distance mean? Decreasing at a increasing rate - use quadratic formula for when distance on airfare is 0.

Set Panel

```
xtset id year
```

Set Panel

```
panel variable: id (strongly balanced)
time variable: year, 1997 to 2000
delta: 1 unit
```

Estimate a Pooled OLS

```
est clear
eststo m1: reg lfare concen ldist ldistsq i.y99 i.y00
```

Source		SS	df	MS	Number of obs	=	4,596
-----+					F(5, 4590)	=	627.18
Model		355.197587	5	71.0395174	Prob > F	=	0.0000
Residual		519.896787	4,590	.113267274	R-squared	=	0.4059
-----+					Adj R-squared	=	0.4052
Total		875.094374	4,595	.190444913	Root MSE	=	.33655

lfare		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
concen		.3609856	.0300677	12.01	0.000	.3020384	.4199327
ldist		-.9018838	.1282905	-7.03	0.000	-1.153395	-.6503727
ldistsq		.1030514	.0097268	10.59	0.000	.0839822	.1221207
1.y99		.0272979	.0121656	2.24	0.025	.0034475	.0511483
1.y00		.0893211	.012169	7.34	0.000	.0654641	.1131782
_cons		6.219743	.4206248	14.79	0.000	5.395116	7.04437

Estimate a Random Effects model

```
eststo m2: xtreg lfare concen ldist ldistsq i.y99 i.y00, re theta
```

Random-effects GLS regression	Number of obs	=	4,596
Group variable: id	Number of groups	=	1,149
R-sq:	Obs per group:		
within = 0.1282	min =		4
between = 0.4179	avg =		4.0
overall = 0.4030	max =		4

```

corr(u_i, X)  = 0 (assumed)
theta         = .83489895

Wald chi2(5)  = 1331.49
Prob > chi2   = 0.0000

```

lfare	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
concen	.216343	.0265686	8.14	0.000	.1642694	.2684166
ldist	-.8544998	.2464622	-3.47	0.001	-1.337557	-.3714428
ldistsq	.0977308	.0186343	5.24	0.000	.0612083	.1342532
1.y99	.0255419	.0038794	6.58	0.000	.0179385	.0331454
1.y00	.0870883	.0038876	22.40	0.000	.0794686	.0947079
_cons	6.232589	.8098919	7.70	0.000	4.64523	7.819948
sigma_u	.31930578					
sigma_e	.1069025					
rho	.89920879	(fraction of variance due to u_i)				

Estimate a FE model

```
eststo m3: xtreg lfare concen ldist ldistsq i.y99 i.y00, fe
```

note: ldist omitted because of collinearity
note: ldistsq omitted because of collinearity

```

Fixed-effects (within) regression      Number of obs   =    4,596
Group variable: id                    Number of groups =    1,149

```

```

R-sq:                                Obs per group:
    within = 0.1286                  min =          4
    between = 0.0576                  avg  =         4.0
    overall = 0.0102                  max  =          4

```

```

corr(u_i, Xb) = -0.2143              F(3,3444)       =    169.47
                                      Prob > F         =    0.0000

```

lfare	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
concen	.177761	.0294665	6.03	0.000	.1199873	.2355346
ldist	0	(omitted)				
ldistsq	0	(omitted)				
1.y99	.0250736	.0038791	6.46	0.000	.017468	.0326791
1.y00	.0864927	.0038892	22.24	0.000	.0788673	.0941181

```

      _cons |   4.959254   .0183174   270.74   0.000   4.92334   4.995168
-----+-----
      sigma_u |  .43441394
      sigma_e |  .1069025
      rho |  .94290028   (fraction of variance due to u_i)
-----+-----
F test that all u_i=0: F(1148, 3444) = 60.09           Prob > F = 0.0000

```

Compare

```
esttab m1 m2 m3, mtitle("OLS" "RE" "FE")
```

Compare

	(1) OLS	(2) RE	(3) FE
concen	0.361*** (12.01)	0.216*** (8.14)	0.178*** (6.03)
ldist	-0.902*** (-7.03)	-0.854*** (-3.47)	0 (.)
ldistsq	0.103*** (10.59)	0.0977*** (5.24)	0 (.)
1.y99	0.0273* (2.24)	0.0255*** (6.58)	0.0251*** (6.46)
1.y00	0.0893*** (7.34)	0.0871*** (22.40)	0.0865*** (22.24)
_cons	6.220*** (14.79)	6.233*** (7.70)	4.959*** (270.74)
N	4596	4596	4596

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Chapter 4

Formatting the display of variables

Formatting data will be more common than you expect. It can be a pain when dealing with numbers in the millions or billions and you lack commas. We can format our data with the *format* command.

4.1 Format numerics

Let's get our survey data and list the first 5 observations for id and income

Let's look at the format of income.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use survey5, clear
describe income
list id income in 1/5
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
(Survey of graduate students)
```

variable name	storage type	display format	value label	variable label
income	float	%9.0g		Income of student

+-----+

```

      | id      income |
      |-----|
1.   | 1      10500.93 |
2.   | 2      45234.13 |
3.   | 3      1284355   |
4.   | 4      124313.5  |
5.   | 5      120102.3  |
      +-----+

```

The format is `%9.0g`. We always have `%` in front of our format and `g` is a general way of displaying incomes using a width of nine digits and decides for us the best way to display the values. `g` means general here.

`%0g` means general - find the best way to show the decimals.

Note: `%g` will change the format to exponent if necessary. Also is usually set to 0 with `g`.

`%f` means fixed - `w` is the width, `d` is the decimals, and `f` means fixed

`%fc` means fixed with commas - `w` is the width, `d` is the decimals, `f` means fixed, and `c` means comma

`%0gc` means general with commas - `w` is the width, setting 0 means Stata will decide the decimals, `g` means general, and `c` means comma

The manual is helpful for formatting: <https://www.stata.com/manuals/dformat.pdf>

Example 1: `format v1 %10.0g` - Width of 10 digits and decimals will be decided.

Example 2: `format v2 %4.1f` - Show 3 digits in `v3` and 1 decimal

Example 3: `format v3 %6.1fc` - Show 4 digits plus the comma plus 1 digit

Let's get more control over the income format and use the `%w.df` format. We want a total of 12 digits with 2 decimals places, which means we have 10 digits on the left side of the “.”

```

format income %12.2f
list income in 1/5

```

```

      |      income |
      |-----|
1.   | 10500.93 |
2.   | 45234.13 |
3.   | 1284354.50 |
4.   | 124313.45 |
5.   | 120102.32 |
      +-----+

```

Notice that we now can see observation #3's decimal places.

If we don't care to see the decimal place (even though it is still there).

```
format income %7.0f
list income in 1/5
```

```

      |   income |
      |-----|
1.   |    10501 |
2.   |    45234 |
3.   |   1284354 |
4.   |    124313 |
5.   |    120102 |
      +-----+
```

We we want to see one decimal place

```
format income %9.1f
list income in 1/5
```

```

      |   income |
      |-----|
1.   |   10500.9 |
2.   |   45234.1 |
3.   |  1284354.5 |
4.   |   124313.5 |
5.   |   120102.3 |
      +-----+
```

Now let's add commas, but we need to add two additional digit widths for the commas and we'll add two decimal places.

```
format income %12.2fc
list income in 1/5
```

```

      |   income |
      |-----|
1.   |  10,500.93 |
2.   |  45,234.13 |
3.   | 1,284,354.50 |
4.   |  124,313.45 |
5.   |  120,102.32 |
      +-----+
```

4.2 Format Strings

Let's use the *format* command with strings.

```
describe kidname
```

variable name	storage type	display format	value label	variable label
kidname	str10	%10s		Name of child

The format is *%10s*, which is a (s)tring of 10 characters wide that is right-justified.

```
list kidname
```

```

      | kidname |
      |-----|
1.   |         |
2.   |   Sally  |
3.   | Catherine|
4.   |         |
5.   | Samuell  |
      |-----|
6.   |         |
7.   |   Robin  |
8.   |         |
      +-----+
```

If we wanted to left-justify the string, we can add a '-' in between % and #s.

```
format kidname %-10s
```

```
list kidname
```

```

      | kidname |
      |-----|
1.   |         |
2.   | Sally    |
3.   | Catherine|
4.   |         |
5.   | Samuell  |
      |-----|
6.   |         |
7.   | Robin    |
8.   |         |
      +-----+
```

4.3 Format dates

Dates in Stata are a bit of a pain, so learning how to format the dates will be helpful in the future.

```
list bdays kbdays
```

	bdays	kbdays
1.	1/24/1961	
2.	4/15/1968	4/15/1995
3.	5/23/1971	8/15/2003
4.	6/25/1973	
5.	9/22/1981	1/12/1999
6.	10/15/1973	
7.	7/1/1977	5/20/1998
8.	8/3/1976	

Our birthdays are in a MM/DD/YYYY format currently. Let's generate a new variable with the *date* function. The *date* function will convert a string that is in a date format into a Stata date, but it still needs to be formatted. The option "MDY" tells Stata that the string is in the Month-Day-Year format and needs to be converted.

```
generate bday = date(bdays, "MDY")
generate kbday = date(kbdays, "MDY")
```

Let's list the days.

```
list bdays bday kbdays kbday
```

	bdays	bday	kbdays	kbday
1.	1/24/1961	389		.
2.	4/15/1968	3027	4/15/1995	12888
3.	5/23/1971	4160	8/15/2003	15932
4.	6/25/1973	4924		.
5.	9/22/1981	7935	1/12/1999	14256
6.	10/15/1973	5036		.
7.	7/1/1977	6391	5/20/1998	14019
8.	8/3/1976	6059		.

The Stata dates are actually stored as the number of days from Jan 1, 1960. This method is convenient for the computer storing and performing date computations, but is difficult for us to read.

Let's use the `%td` format - for example 01Jan2000

```
format bday %td
list bdays bday kbdays kbdays
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	24jan1961		.
2.	4/15/1968	15apr1968	4/15/1995	12888
3.	5/23/1971	23may1971	8/15/2003	15932
4.	6/25/1973	25jun1973		.
5.	9/22/1981	22sep1981	1/12/1999	14256
6.	10/15/1973	15oct1973		.
7.	7/1/1977	01jul1977	5/20/1998	14019
8.	8/3/1976	03aug1976		.

Let's use the `%tdNN/DD/YY` format...*NN* is used for 01-12 and *nn* is for 1-12, *DD* for the day 01-31, and *YY* is for the last two digits of the year.

```
format bday %tdNN/DD/YY
list bdays bday kbdays kbdays
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	01/24/61		.
2.	4/15/1968	04/15/68	4/15/1995	12888
3.	5/23/1971	05/23/71	8/15/2003	15932
4.	6/25/1973	06/25/73		.
5.	9/22/1981	09/22/81	1/12/1999	14256
6.	10/15/1973	10/15/73		.
7.	7/1/1977	07/01/77	5/20/1998	14019
8.	8/3/1976	08/03/76		.

Mon is Jan-Dec, and *Month* is January-December.

```
format bday %tdMonth/DD/YY
list bdays bday kbdays kbdays
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	January/24/61		.
2.	4/15/1968	April/15/68	4/15/1995	12888
3.	5/23/1971	May/23/71	8/15/2003	15932
4.	6/25/1973	June/25/73		.
5.	9/22/1981	September/22/81	1/12/1999	14256
6.	10/15/1973	October/15/73		.
7.	7/1/1977	July/01/77	5/20/1998	14019
8.	8/3/1976	August/03/76		.

We can use a standard Month DD, YYYY with the format `%tdMonth_DD,CCYY`. Where *Month* is the full name of the month, *DD* is our days in digits, and *CC* is Century, such as 19- and 20- and *YY* is 2-digit year, such as -88, -97

```
format bday %tdMonth_DD,CCYY
list bdays bday kbdays kbdays
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	January 24,1961		.
2.	4/15/1968	April 15,1968	4/15/1995	12888
3.	5/23/1971	May 23,1971	8/15/2003	15932
4.	6/25/1973	June 25,1973		.
5.	9/22/1981	September 22,1981	1/12/1999	14256
6.	10/15/1973	October 15,1973		.
7.	7/1/1977	July 01,1977	5/20/1998	14019
8.	8/3/1976	August 03,1976		.

Let's use a standard format, but don't use YYYY - it just repeats the 2-digit year twice.

```
format bday %tdNN/DD/YYYY
list bdays bday kbdays kbdays
```

	bdays	bday	kbdays	kbdays
--	-------	------	--------	--------

1.	1/24/1961	01/24/6161	.
2.	4/15/1968	04/15/6868	4/15/1995 12888
3.	5/23/1971	05/23/7171	8/15/2003 15932
4.	6/25/1973	06/25/7373	.
5.	9/22/1981	09/22/8181	1/12/1999 14256
6.	10/15/1973	10/15/7373	.
7.	7/1/1977	07/01/7777	5/20/1998 14019
8.	8/3/1976	08/03/7676	.

Use %tdNN/DD/CCYY instead for the desired result.

```
format bday %tdNN/DD/CCYY
list bdays bday kbdays kbdays

label variable bday "Date of birth of student"
label variable kbdays "Date of birth of child"
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	01/24/1961	.	
2.	4/15/1968	04/15/1968	4/15/1995 12888	
3.	5/23/1971	05/23/1971	8/15/2003 15932	
4.	6/25/1973	06/25/1973	.	
5.	9/22/1981	09/22/1981	1/12/1999 14256	
6.	10/15/1973	10/15/1973	.	
7.	7/1/1977	07/01/1977	5/20/1998 14019	
8.	8/3/1976	08/03/1976	.	

bday and bdays are redundant and we'll only keep one.

```
drop bday kbdays
save survey6, replace
```

Chapter 5

Changing the order of variables in a dataset

I personally find reordering the order of variables with the *order* command to be useful. This is especially true when working with panel data. I like to order the panel data to have the cross-sectional unit first, such as personal id, firm id, etc.first and then have the time period second, so we have our N and T next to one another.

Let's pull our survey of graduate students and describe our dataset

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"  
use survey6, clear  
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
(Survey of graduate students)
```

```
Contains data from survey6.dta
```

obs:	8	Survey of graduate students
vars:	9	11 Mar 2024 14:40
size:	416	(_dta has notes)

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g	mf	Gender of student

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```

race          float   %19.0g   racelab * Race of student
havechild     float   %18.0g   havelab * Given birth to a child?
ksex          float   %15.0g   mfkid  * Sex of child
bdays        str10    %10s      Birthday of student
income        float   %12.2fc   Income of student
kidname       str10    %-10s     Name of child
kday          double   %td

* indicated variables have notes

```

Sorted by:

We might want to group our variables with similar types of variables. This can be helpful when you have a large dataset with hundreds of variables, such as the CPS.

```

order id gender race bday income havechild
describe

```

```

Contains data from survey6.dta
  obs:          8          Survey of graduate students
  vars:          9          11 Mar 2024 14:40
  size:         416         (_dta has notes)

```

```

      storage  display  value
variable name  type    format  label    variable label
-----
id            float    %9.0g      mf      Unique identification variable
gender        float    %9.0g      mf      Gender of student
race          float    %19.0g   racelab * Race of student
bdays        str10    %10s      Birthday of student
income        float    %12.2fc   Income of student
havechild     float    %18.0g   havelab * Given birth to a child?
ksex          float    %15.0g   mfkid  * Sex of child
kidname       str10    %-10s     Name of child
kday          double   %td

* indicated variables have notes

```

Sorted by:

The variables that we leave off will remain in the same order as before after the new variables are moved to the left.

With the *before* option, we can move variable(s) before a defined variable. Let's move kidname before ksex

```
order kidname, before(ksex)
describe
```

Contains data from survey6.dta

```
obs:      8                      Survey of graduate students
vars:     9                      11 Mar 2024 14:40
size:    416                     (_dta has notes)
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g	mf	Gender of student
race	float	%19.0g	racelab	* Race of student
bdays	str10	%10s		Birthday of student
income	float	%12.2fc		Income of student
havechild	float	%18.0g	havelab	* Given birth to a child?
kidname	str10	%-10s		Name of child
ksex	float	%15.0g	mfkid	* Sex of child
kday	double	%td		

* indicated variables have notes

Sorted by:

We can move newly created variables with the before and after options with the generate command

```
generate STUDENTVARS = ., before(gender)
generate KIDSVARS = ., after(havechild)
describe
```

(8 missing values generated)

(8 missing values generated)

Contains data from survey6.dta

```
obs:      8                      Survey of graduate students
vars:    11                      11 Mar 2024 14:40
size:   544                     (_dta has notes)
```

variable name	storage type	display format	value label	variable label
---------------	--------------	----------------	-------------	----------------

```

id          float   %9.0g          Unique identification variable
STUDENTVARS double   %10.0g
gender      float   %9.0g      mf      Gender of student
race        float   %19.0g      racelab * Race of student
bdays      str10   %10s          Birthday of student
income      float   %12.2fc          Income of student
havechild   float   %18.0g      havelab * Given birth to a child?
KIDSVARS    double   %10.0g
kidname     str10   %-10s          Name of child
ksex        float   %15.0g      mfkid   * Sex of child
kday        double   %td

```

* indicated variables have notes

Sorted by:

Note: Dataset has changed since last saved.

5.1 Practice

Let's bring in the CPS: <https://www.census.gov/data/datasets/time-series/demo/cps/cps-basic.html>

```

cd "/Users/Sam/Desktop/Econ 645/Data/CPS/"
use jul25pub.dta, clear

```

1. Generate a new variable from `pemlr` called `employed` where `employed = 1` if the individual is employed (present or absent) and `employed = 0` if the individual is unemployed. The value should be missing if the individual is not in the labor force.
2. Label the variable "Currently employed".
3. Label the values for 0 "Not employed" 1 "Employed" . "Not in the Labor Force".
4. Move the variable after `pemlr`.
5. Generate a date that appends `hrmonth` (month of interview), the string "12", and the `hryear4` (year of interview). We use 12 because the week of the 12th is the reference period.
6. Now format the date so it is like 07/12/2025